

Introduction to Cell Arrays

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Part 1: What Are Cell Arrays?

Cell arrays are containers that can hold data of varying types and sizes. Unlike numeric vectors and matrices, cell arrays can store strings, numbers, other arrays, or even complex objects in the same structure.

Create a cell array

```
C = {'Hello', 42, [1, 2, 3; 4, 5, 6], pi};
disp(C);
```

{'Hello'} {[42]} {2x3 double} {[3.1416]}

Access elements in a cell array

First element (a string)

```
C{1}
```

ans =
'Hello'

Second element (a number)

```
C{2}
```

ans = 42

Third element (a matrix):

```
C{3}
```

ans = 2x3
1 2 3
4 5 6

Fourth element (a number):

```
C{4}
```

ans = 3.1416

Part 2: Differences Between Cell Arrays and Numeric Vectors/Matrices

Numeric Arrays:

- Store only numeric data of the same size and type.
- Perform mathematical operations directly.

Cell Arrays:

- Can store mixed data types.
- Use curly braces {} to access data (unlike parentheses for numeric arrays).

Example: Numeric Vector

```
vec = [1, 2, 3]
```

```
vec = 1x3
      1      2      3
```

Example: Cell Array with Mixed Types

```
cell_vec = {1, 'two', [3, 3, 3]};
disp(cell_vec);
```

```
{[1]}      {'two'}      {[3 3 3]}
```

Part 3: Indexing Cell Arrays

Access data using {} to extract the content

```
value = C{2}
```

```
value = 42
```

Access data using () to extract a cell (returns a cell array)

```
cell_value = C(2)
```

```
cell_value = 1x1 cell array
    {[42]}
```

Modify elements in a cell array

```
C{2} = 100; % Change the value in the second cell
disp(C);
```

```
{'Hello'}      {[100]}      {2x3 double}      {[3.1416]}
```

Part 4: Combining Cell Arrays

Concatenate cell arrays

```
C2 = {'New', 99};
C_combined = [C, C2];
disp(C_combined);
```

```
{'Hello'}      {[100]}      {2x3 double}      {[3.1416]}      {'New'}      {[99]}
```

Part 5: More Examples

Example 1: Storing Strings and Numbers Together

```
data = {'Alice', 85; 'Bob', 92; 'Charlie', 78};  
disp(data);
```

```
    {'Alice' }    {[85]}  
    {'Bob'   }    {[92]}  
    {'Charlie'}   {[78]}
```

Access specific elements

```
data{1, 1}
```

```
ans =  
'Alice'
```

```
data{2, 2}
```

```
ans = 92
```

Example 2: Storing Matrices of Different Sizes

```
matrices = {rand(2, 2), rand(3, 3), rand(4, 4)};  
disp(matrices);
```

```
    {2x2 double}    {3x3 double}    {4x4 double}
```

Access and manipulate specific matrix

```
matrix_2x2 = matrices{1};  
disp(matrix_2x2);
```

```
    0.1829    0.8865  
    0.2399    0.0287
```